

Yep. For testing **our actual pipeline**, don't use a trivial proposal. We want something that deliberately contains **similar historical work + a genuine technical advancement + weaknesses + budget/rule issues**, so we can see whether the tribunal actually reasons or just says "novel."

Here's a good synthetic test proposal.

Sample R&D Proposal

Title

Predictive AI-Based Methane Risk Forecasting and Early-Warning System for Underground Coal Mines

Proposal Type

R&D / Technology Development

Domain

Mining Safety / Artificial Intelligence / IoT

Duration

24 months

Requested Funding

₹2.85 Crore

Organization

Synthetic Demo Organization — not an actual CMPDI proposal

1. Problem Statement

Methane accumulation is a major safety risk in underground coal mines. Existing methane monitoring systems primarily measure the current concentration of methane at predefined locations and trigger alarms when measured concentrations exceed configured thresholds.

While these systems provide real-time detection, they generally provide limited warning about **future methane concentration increases**.

A sudden increase in methane concentration may therefore leave mine personnel with limited time to take preventive action.

This project proposes an AI-based predictive system capable of forecasting methane concentration and estimating the probability of a hazardous methane event **15–30 minutes before it occurs**.

2. Proposed Solution

The proposed system will combine:

- Fixed methane sensors
- Temperature sensors
- Humidity sensors
- Ventilation parameters
- Airflow measurements
- Historical methane measurements
- Equipment operating conditions

An AI forecasting model will process these signals and predict methane concentration for the next 15, 30 and 60 minutes.

The system will generate:

NORMAL

→ **ELEVATED RISK**

→ **HIGH RISK**

alerts.

The system will also provide an explanation of the major factors contributing to the predicted risk.

3. Novelty Claim

The applicant claims that existing mine safety systems primarily focus on **detecting methane concentration after or during its increase**, whereas the proposed system

focuses on **predicting future methane concentration and risk before a dangerous threshold is reached.**

The proposal claims that this transition from:

Reactive Detection → Predictive Risk Forecasting

represents a significant advancement in mine safety technology.

4. Technical Methodology

Stage 1 — Data Collection

Collect historical data from underground mine sensors.

Target dataset:

- 2 years of historical data
- 20 methane sensors
- 10 airflow sensors
- 10 environmental sensors

Expected dataset size:

Approximately **50 million sensor observations.**

Stage 2 — Data Processing

The system will:

- remove sensor noise
 - identify missing readings
 - detect sensor failures
 - normalize sensor values
 - synchronize measurements from different sensors
-

Stage 3 — AI Model

The initial model will use:

LSTM-based time-series forecasting

A second model based on **Temporal Fusion Transformer (TFT)** will be evaluated during the project.

The models will predict:

Methane concentration after 15 minutes

Methane concentration after 30 minutes

Methane concentration after 60 minutes

5. Expected Accuracy

The project targets:

≥90% prediction accuracy

and:

≥85% hazardous-event prediction recall.

However, the proposal currently does **not clearly define** whether "90% accuracy" means:

- classification accuracy
 - MAE
 - RMSE
 - precision
 - recall
 - F1 score
-

6. Baseline

The proposal states:

Existing sensor-based methane monitoring systems will be used as the baseline.

However, it does not specify:

- exact baseline system
 - baseline model
 - baseline prediction accuracy
 - historical false-positive rate
 - historical false-negative rate
-

7. Validation Plan

The proposal proposes:

- 70% training data
- 15% validation data
- 15% testing data

The model will be evaluated using historical sensor data.

A field trial will subsequently be performed in one underground mining site.

The proposal does not clearly explain how the field trial will establish that the AI system improves safety compared with the existing monitoring system.

8. Expected Outcomes

The project expects to deliver:

1. AI-based methane forecasting model.
 2. Real-time methane risk dashboard.
 3. Early-warning system.
 4. Mobile alerts for authorized personnel.
 5. API for integration with mine monitoring systems.
 6. Model performance monitoring system.
 7. Prototype deployment in an underground mine.
-

9. Expected Impact

The proposed system is expected to:

- provide earlier warning of methane accumulation
- reduce reaction time
- support preventive evacuation
- improve mine safety
- reduce false alarms
- support mine safety personnel in decision-making

The proposal estimates:

20% reduction in methane-related emergency response time.

However, the proposal does not provide sufficient evidence explaining how this number was derived.

10. Budget

Component	Amount
Sensor integration	₹35 lakh
Computing infrastructure	₹40 lakh
AI model development	₹60 lakh
Software development	₹35 lakh
Field deployment	₹45 lakh
Personnel	₹50 lakh
Contingency	₹20 lakh
Total	₹2.85 Crore

11. Timeline

Months 1–6

Data collection and preprocessing.

Months 7–12

Model development.

Months 13–18

Model testing and optimization.

Months 19–21

Field deployment.

Months 22–24

Evaluation and final prototype.

12. Risks

Potential risks include:

- sensor failures
 - insufficient historical data
 - changing mine conditions
 - model drift
 - false alarms
 - missed hazardous events
 - network connectivity failures
 - difficulty integrating with existing monitoring infrastructure
-

13. Limitations

The proposal currently does not clearly specify:

- minimum required dataset quality
 - handling of previously unseen mine conditions
 - model retraining frequency
 - explainability methodology
 - cybersecurity architecture
 - fail-safe behaviour when the AI model fails
 - precise definition of "hazardous event"
-

14. What Makes This Proposal Interesting for Our Tribunal

This is intentionally designed to stress-test the system.

Your RAG should potentially find historical work involving:

methane monitoring / methane detection / sensor-based mine safety

The vector similarity could therefore be **very high**.

A dumb system would say:

Similarity = 0.89

Therefore:

LOW NOVELTY

DUPLICATE

REJECT

Our system should instead reason:

Historical capability:

Real-time methane DETECTION

Current proposal:

Future methane PREDICTION

Similarity:

HIGH

Semantic difference:

SIGNIFICANT

Potential advancement:

DETECTION → PREDICTION

Novelty:

Potentially PROGRESSIVE

BUT...

Then the **Critic** should attack the weak parts:

- "90% accuracy" is undefined.
- No strong baseline is specified.
- Field validation methodology is weak.
- 20% emergency-response improvement has no supporting evidence.
- Safety-critical failure handling is insufficient.
- Dataset assumptions are not adequately justified.

And the **Advocate** should defend it:

- Prediction is materially different from detection.
- Historical monitoring systems provide the data but don't necessarily provide forecasting.
- 15–30 minute advance warning could enable preventive action.
- The proposal includes field deployment.
- It evaluates multiple forecasting architectures.

Then the **Judge** should reach its own conclusion.

For example, a reasonable result might be:

REVIEW / CONDITIONAL APPROVAL

rather than blindly approving or rejecting it.

That's exactly the kind of case we want for testing because it forces the system to distinguish **high textual similarity from actual technical duplication**.